Market Regime Discovery in FAANG Stocks Using Feature-Engineered K-Means Clustering

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***Abstract*—Stock market behavior is difficult to interpret because price movements are noisy, highly volatile, and shaped by multiple interacting factors. This study presents a feature-engineered clustering framework for analyzing FAANG stock data through unsupervised learning. Rather than relying on raw price values alone, the workflow derives technical indicators such as daily return, volatility, RSI, and MACD, then standardizes these features before applying K-Means clustering to reveal hidden market structure. The implementation follows a compact analytical pipeline that computes engineered indicators from historical stock records and groups similar market states into distinct regimes, as reflected in the provided code and paper draft . The resulting clusters support an interpretable division of market behavior into bullish, neutral, and bearish conditions, showing that feature-driven unsupervised learning can provide a practical and academically strong way to study financial market dynamics.**

***Index Terms*—FAANG dataset, K-Means clustering, Market behavior analysis, Technical indicators, Unsupervised machine learning, Financial analytics**

I. Introduction

Financial markets are really complicated because so many things affect how prices move. We're talking about things like how the economy is doing, how well companies are performing, what investors are thinking, and big events that happen around the world. All these things together create patterns, but they can be hard to understand because they're messy and noisy. When we just look at the basic price data, we tend to focus on what's happening right now, which makes it tough to figure out what's really a trend and what's just random. This means we might miss important patterns in how the market behaves.

To better understand how the market works, we often change stock price data into more useful information that shows how the market behaves over time. We use things like return, volatility, and momentum to get a clearer picture of what's happening. For instance, returns show us how prices change from one time to another, volatility tells us how unsure we are, and momentum shows us if a trend is getting stronger or weaker. When we put all this information together in a neat dataset and make sure everything is on the same scale, we can compare things more easily and make a more balanced analysis. This helps get rid of unnecessary information and makes it easier to see what's really going on. By doing this, we can spot important patterns that might be hard to see otherwise.

When it comes to financial data, we don't usually have clear labels for different market situations. That's where unsupervised learning comes in - it's really helpful in this case. One technique that's used is clustering, which groups similar data points together. This allows us to find natural patterns in the data without making any assumptions beforehand. By doing this, we can divide the market into different states, like when it's stable, growing, or when there's a lot of risk involved. This way, we can get a better understanding of what's going on in the market.

This framework isn't about guessing what prices will be in the future. It's really about seeing how the market works and changes over time. By finding patterns in the market, analysts can understand what's going on and make better decisions. They can see how the market is organized and how it might change, which helps them make more informed choices.

II. Dataset Description

The data set used in this project consists of historical stock market data from FAANG companies, namely Facebook (Meta), Apple, Amazon, Netflix, and Google (Alphabet). These companies were selected because they represent some of the most influential and actively traded technology stocks in the market, making them highly suitable for behavior-based financial analysis.

The data set contains daily stock market records that are commonly used for market analysis and financial modeling. The original stock attributes available in the dataset are: Open Price, High Price, Low Price, Close Price, Trading Volume.

Although these raw stock attributes provide important market information, they do not directly reveal a hidden behavioral structure in a form suitable for clustering. Therefore, the data set was further transformed into a more informative feature-rich representation using selected technical and statistical indicators.

These transformed features were used as the primary input for clustering because they better capture momentum, trend, and price variability compared to raw stock prices alone. This makes the data set more appropriate for unsupervised learning and stock market behavior analysis.

TABLE I
SUMMARY OF THE FAANG STOCK DATASET

Parameter	Description
Dataset Type	Historical Stock Market Data
Companies Covered	Facebook, Apple, Amazon, Netflix, Google
Sector	Technology
Raw Features	Open, High, Low, Close, Volume
Learning Type	Unsupervised Learning
Model Used	K-Means Clustering
Purpose	Stock Market Behavior Analysis

A. Dataset Summary

III. PROPOSED MODEL

The proposed model is developed to identify hidden patterns in stock market behavior using an unsupervised learning framework. In contrast to conventional stock market approaches that primarily emphasize price forecasting, this work focuses on market regime discovery through clustering-based behavioral analysis.

The framework converts raw historical stock market data into a more informative feature space by incorporating technical indicators and engineered financial attributes. K-Means clustering is then applied to group structurally similar market conditions based on their statistical and behavioral characteristics. This enables the identification of meaningful and interpretable market regimes, such as bullish, stable, and bearish conditions.

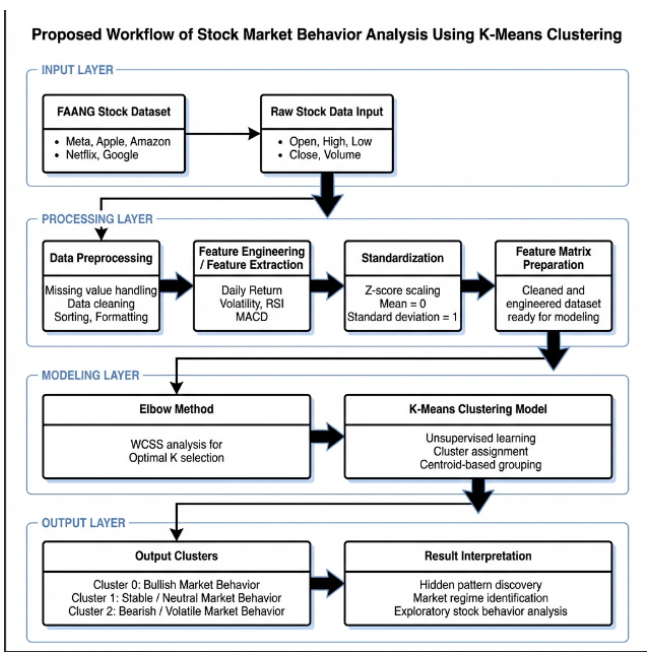

Fig. 1. Proposed workflow of stock market behavior analysis using K-Means clustering

The overall workflow consists of the following stages: Feature selection and engineering, Feature extraction from stock data, Data standardization, K-Means clustering, Cluster evaluation.

A. Features of the Proposed Model

Raw stock prices alone provide limited insight into underlying market behavior, as they primarily reflect absolute values without capturing dynamic changes such as trend, risk, or momentum. To address this, the model combines primary stock attributes—Open, High, Low, Close, and Volume—with engineered technical indicators. Daily Return measures price change over time, capturing gain or loss patterns. Volatility reflects the degree of price fluctuation, indicating market risk. RSI (Relative Strength Index) identifies momentum and overbought or oversold conditions, while MACD (Moving Average Convergence Divergence) highlights trend direction and strength. This combination of features creates a more informative representation, enabling more effective clustering and improved understanding of market behavior.

B. Feature Extraction

Feature extraction is an integral part of the proposed model. It is the process by which the historical data related to the stock market is converted into useful numerical data. Instead of directly using the price-related columns, the model creates other relevant features. These features describe the behavior of the stock market. Studies related to the analysis of the stock market describe the transformed data to effectively describe the behavior related to the trends, risk, and momentum [13], [15].The features extracted by the model can be described as follows:

Daily Return: It is the percentage change in the closing price over the course of two successive days. It is the change in the closing price. It is used to determine the gain or loss in the value of the stocks.

Volatility: It is the standard deviation of the returns over a certain period. It measures the risk involved in the stocks. High volatility indicates high risk in the market.

RSI (Relative Strength Index): It is the momentum indicator which is used to determine the overbought and oversold conditions. It is the measure to understand the short-term market behavior.

MACD (Moving Average Convergence Divergence): It is the measure to understand the trends and the momentum. It is the convergence and divergence of the moving averages.

C. Standardization

The selected features are then standardized before applying the clustering algorithm, to ensure that all features contribute equally to the clustering process. This is done because, in the dataset, some features have different ranges of numerical values. For example, Volume might be in millions, whereas RSI will be within a range of 0 to 100. This will affect the clustering because, in K-means, distance calculations are done, and different ranges of values will affect clustering.

As mentioned in the Scikit-learn library documentation and other clustering algorithm references, distance-based

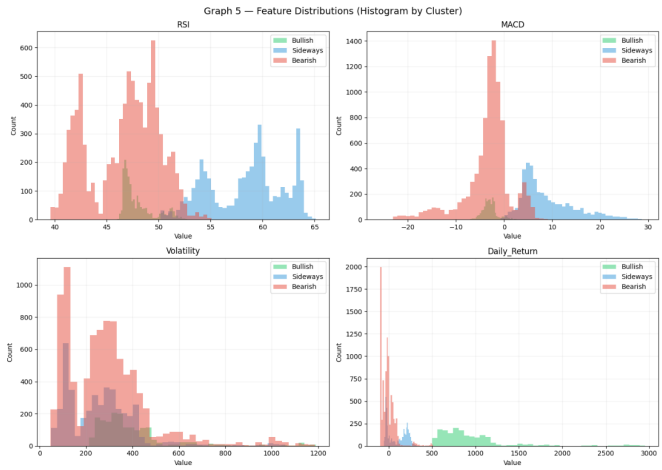

Fig. 2. Feature distribution of engineered stock market indicators across bullish, sideways, and bearish market clusters.

clustering is scale-dependent.

To avoid such a problem, standardization is applied to the dataset, also known as z-score scaling, using the following equation:

$$Z = \frac{X - \mu}{\sigma} \tag{1}$$

Where:

X = value of a feature
μ = mean of a feature
σ = standard deviation of a feature

As a result of applying standardization to the dataset, we will have: each feature's mean = 0 each feature's standard deviation = 1 This ensures that no feature affects clustering more than another, thus making clustering more fair and quality-oriented. This approach was also adopted in a recent clustering algorithm-based finance-oriented research paper [12], [15].

D. Model Description

K-Means clustering is a suitable method for stock market analysis because financial datasets usually do not contain explicit labels such as bullish or bearish conditions. Instead of relying on predefined categories, the algorithm finds patterns directly from the data by grouping similar observations together.

The process is iterative. First, the number of clusters KKK is selected and initial cluster centers are chosen. Each data point is then assigned to the nearest center using a distance measure such as Euclidean distance. After that, the centers are recalculated as the mean of the assigned points. These steps repeat until the clusters no longer change significantly.

To choose the best value of KKK, the Elbow Method is used by examining the within-cluster sum of squares (WCSS). The point where the decrease in WCSS begins to slow down indicates a suitable balance between simplicity and performance. In this study, K=3K=3K=3 is selected, representing three market regimes: bullish, neutral or stable, and bearish or volatile.

Overall, the model groups feature-engineered financial data into meaningful clusters, helping reveal different market states and improve understanding of market behavior without attempting price prediction.

E. Performance Evaluation

Since K-Means is an unsupervised learning algorithm, supervised evaluation metrics such as accuracy, precision, recall, and F1-score are not applicable. Therefore, the proposed model is evaluated using clustering quality measures that assess how well the data points are grouped. The major evaluation criteria are: (i) WCSS (Within-Cluster Sum of Squares)

WCSS measures the squared distance between data points and their respective cluster centroids. Lower WCSS indicates more compact clusters. (ii) Elbow Method

The Elbow Method is used to determine the optimal number of clusters by analyzing the rate of decrease in WCSS for different values of K. (iii) Cluster Compactness

Cluster compactness indicates how closely related the data points are within the same cluster. (iv) Cluster Separation

Cluster separation measures how distinctly different one cluster is from another. (v) Interpretability

An effective clustering model should produce meaningful groups that can be interpreted as stock market regimes such as bullish, stable, and bearish behavior.

F. Figures and Tables

TABLE II
DATASET SUMMARY STATISTICS

Feature	Mean	Max
Open	144.09	782.98
High	145.74	795.06
Close	144.15	784.06
Volume	1.17e8	1.65e9
RSI	55.63	99.42
Daily Return	0.0013	0.113
Volatility	0.0185	0.118

The dataset comprises key features including Open, High, Low, Close prices, Volume, RSI, and Daily Return. Price features exhibit similar averages and strong correlations, indicating limited intra-day variation and shared market behavior. Volume shows significantly higher scale and variability, reflecting trading activity.

RSI remains within its expected range and has weak correlation with price features, capturing momentum. Daily Returns are small, suggesting gradual price changes. Due to differences in feature scales, standardization is required before clustering. The combination of correlated price features and relatively independent Volume supports effective identification of market patterns. The correlation matrix shows that Open, High, Low, and Close prices

TABLE III
FEATURE CORRELATION MATRIX (PARTIAL)

Feature	Open	Volume	RSI
Open	1.00	-0.41	-0.02
Volume	-0.41	1.00	-0.22
RSI	-0.02	-0.22	1.00

are strongly related, indicating they move together and represent similar aspects of stock price behavior within a trading period. In contrast, trading Volume often shows weak or even negative correlation with price features, suggesting that higher trading activity does not always result in higher prices.
RSI also has a weak relationship with price features, as it reflects momentum rather than absolute price levels. This mix of strongly correlated and relatively independent features provides a balanced dataset, helping clustering methods identify distinct patterns and better understand different market behavior states.

RESULTS

The proposed K-Means clustering framework was applied to the preprocessed and standardized FAANG stock dataset to identify underlying market behavior patterns. The clustering results demonstrate that the selected technical indicators are effective in capturing structural differences in stock market conditions, leading to meaningful separation of data into distinct behavioral groups. Since the model is based on unsupervised learning, the results are evaluated using clustering quality measures such as within-cluster compactness (WCSS) and visual separability of clusters, rather than traditional classification metrics.

Elbow Method Analysis

The Elbow Method was used to determine the optimal number of clusters K. This method evaluates clustering performance by analyzing the variation of inertia (Within-Cluster Sum of Squares) across different values of K. Figure 3 shows that the WCSS decreases rapidly as the number of clusters increases from 1 to 3, indicating significant improvement in cluster compactness. Beyond $K = 3$, the rate of decrease in WCSS becomes considerably smaller, suggesting diminishing returns in clustering performance.

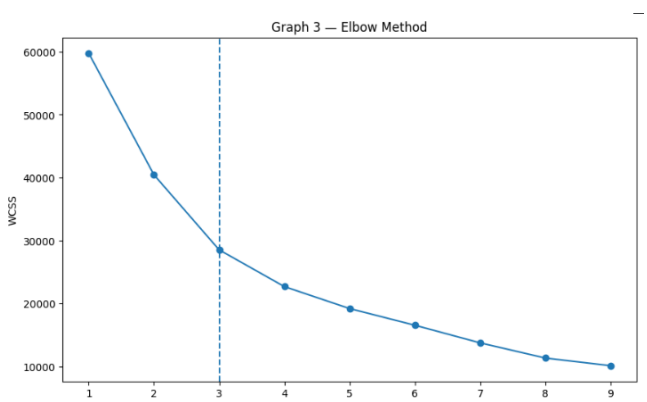

Fig. 3. Elbow method used to determine the optimal number of clusters for K-Means clustering.

Observations

A sharp reduction in WCSS is observed for $K = 1$ to $K = 3$, indicating that additional clusters significantly improve data representation in this range. For $K > 3$, the reduction in WCSS becomes gradual, implying that additional clusters do not provide substantial improvement. The curve exhibits a clear "elbow" at $K = 3$, which indicates the most suitable number of clusters for the dataset.

Inference

Based on the Elbow Method analysis, the optimal number of clusters is determined as $K = 3$. This result aligns with the presence of three distinct market regimes within the dataset. The clustering output confirms that the stock market behavior can be effectively categorized into three meaningful states: Bullish market behavior, Stable / neutral market behavior and Bearish / volatile market behavior. This validates that the proposed feature-engineered representation successfully captures key characteristics of market dynamics and enables interpretable segmentation of stock market conditions [12], [13].

IV. K-MEANS CLUSTERING AND CLUSTER INTERPRETATION

The optimal number of clusters was selected using the Elbow Method, where a sharp drop in WCSS up to K=3K=3K=3 indicated the best balance between simplicity and accuracy. K-Means then groups data points based on similarity rather than predefined labels, making it suitable for unlabeled financial data.
The model uses features such as Daily Return, Volatility, RSI, and MACD, which capture key aspects of market behavior including returns, risk, momentum, and trend direction. Clustering in this feature space helps uncover hidden patterns in historical data, revealing distinct market states and improving understanding of underlying market dynamics.

A. Cluster Visualization Analysis

The clustering visualization shows how stock market data separates into three distinct groups when projected onto feature pairs such as RSI vs. MACD and Daily Return vs. MACD. These plots make the clustering behavior easier to interpret because they place momentum and return-based indicators in the same space, allowing market states to emerge visually rather than being imposed in advance. Figure 4 presents the clustering results using two feature combinations: RSI vs MACD and Daily Return vs MACD. It can be observed that the data points are grouped into three visually distinct clusters, each

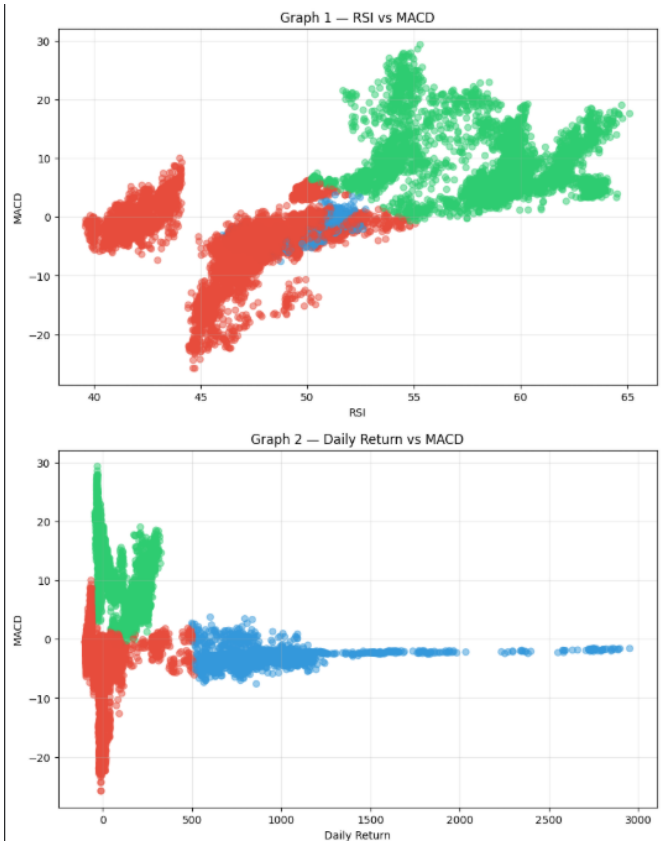

Fig. 4. Cluster visualization using RSI vs MACD and Daily Return vs MACD showing separation of bullish, sideways, and bearish market conditions.

representing a different stock market condition.

In the project code, these features were computed from historical stock prices, standardized, and then clustered with K-Means into three groups . The paper likewise describes these clusters as meaningful market regimes rather than arbitrary partitions .

B. Observations

Several observations stand out. First, the data forms three major clusters, which supports the idea that the market can be segmented into a small number of recurring behavioral states. Second, some overlap is visible, which is expected because financial markets are noisy and do not always separate into perfectly clean groups. Even so, the selected features provide enough structure to distinguish the clusters in a useful way. The scatter plots in the paper show that the combinations of RSI, MACD, and Daily Return produce recognizable separation across bullish, sideways, and bearish regions .

C. Interpretation of Clusters

Cluster 0 represents bullish market behavior. It is associated with higher daily returns, stronger upward momentum, and comparatively stable conditions. Cluster 1 reflects stable or neutral behavior, where returns remain moderate, volatility is balanced, and price movement tends to be sideways. Cluster 2 corresponds to bearish or volatile behavior, marked by lower returns, higher instability, and weaker or negative trend direction.

Overall, the results show that feature engineering, standardization, and K-Means clustering work together to reveal hidden market structure. By focusing on return, volatility, momentum, and trend strength, the model separates historical stock data into interpretable regimes and exposes patterns that are not obvious in raw prices alone .

V. Performance Evaluation

A. Feature Profile Analysis

The feature profile heatmap shows that each cluster has a distinct average pattern across daily return, volatility, RSI, and MACD, confirming that the clusters are not random groupings but statistically meaningful market regimes. Figure 5 presents a heatmap of the average feature values

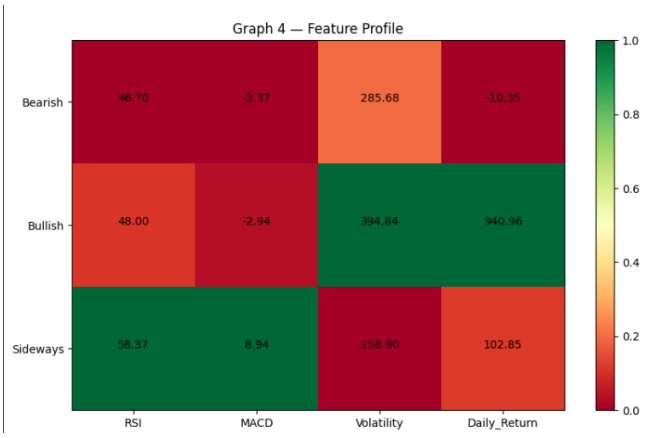

Fig. 5. Feature profile heatmap showing average feature values across different market clusters.

for each cluster. It can be observed that each cluster exhibits a distinct pattern across the selected features. The bullish cluster typically shows higher daily returns and stronger positive momentum, with features suggesting upward movement and relatively favorable conditions. The sideways or neutral cluster is marked by moderate returns, balanced volatility, and feature values that reflect steady but less directional behavior.

The bearish cluster shows lower daily returns, higher volatility, and weaker or negative MACD values, indicating decline and instability. RSI also helps separate these groups by reflecting momentum differences rather than price level alone. Together, these contrasts show that the clustering method captures real structure in the data and separates market behavior in a clear, interpretable way.

B. Discussion

The results of grouping similar data points together show that our approach is good at finding the basic patterns in the stock market. Instead of trying to predict what will happen, we focused on finding patterns in the data. This

way, we can see how the market behaves in different situations. We also used special tools to make the data more useful, which helped us create clear and easy-to-understand groups of data. This makes it easier to see how the market is working.

Looking at the results, it seems that the Elbow Method points to three as the best number of clusters, which means we have three main market regimes. When we visualize the data, we can see that these clusters are pretty distinct, which suggests that the features we've chosen are doing a good job of telling different market conditions apart. It's also clear that using technical indicators like return, volatility, and momentum is crucial in making the clusters more defined, as they capture the key characteristics of the financial markets.

This is important because it helps us understand what's driving the markets and make more informed decisions. By including these indicators, we can get a better sense of what's going on and make more accurate predictions. Overall, the analysis suggests that our approach is effective in identifying the underlying market regimes, which can be useful in a variety of applications, from risk management to portfolio optimization.

C. Limitations

Despite its effectiveness, the proposed framework has certain limitations. The K-Means algorithm assumes that clusters are formed around centroids and are relatively compact in shape, which may not always reflect the complex and non-linear structure of financial market data. In addition, stock market behavior is naturally noisy and dynamic, which can reduce the clarity of cluster boundaries.

Another limitation is that the proposed model relies only on historical stock prices and technical indicators. It does not incorporate external variables such as macroeconomic conditions, market news, investor sentiment, or global events, all of which can significantly influence stock market behavior.

Therefore, the results of this work should be interpreted as an exploratory analysis of market regimes rather than a complete representation of all real-world market influences [14], [15].

VI. Conclusion

This study presented a feature-driven unsupervised learning framework for stock market behavior analysis using FAANG stock data. Instead of relying only on raw stock prices, the proposed approach transformed the historical stock records into more informative technical features such as Daily Return, Volatility, RSI, and MACD. These engineered features provided a more meaningful representation of market behavior for clustering analysis. The K-Means clustering algorithm was applied after preprocessing, feature engineering, normalization, and optimal cluster selection using the Elbow Method. The clustering results showed that the stock market data could be effectively grouped into three meaningful market states: bullish, stable/neutral, and bearish/volatile. The obtained clusters were further supported through visual analysis, feature distributions, boxplots, scatter plots, and feature profile heatmaps. The results demonstrate that unsupervised learning can be successfully used to discover hidden structural patterns in stock market data without relying on predefined labels. The proposed framework improves the interpretability of financial market behavior and provides a structured approach for exploratory market regime analysis. Although the present work focuses on clustering-based market understanding rather than direct price prediction, it establishes a strong foundation for future research. In future work, this framework can be extended by integrating additional technical indicators, deep learning models, hybrid clustering approaches, or real-time stock market analysis systems.

References

[1] J. MacQueen, "Some methods for classification and analysis of multivariate observations," in *Proc. 5th Berkeley Symp. Mathematical Statistics and Probability*, Berkeley, CA, USA, 1967, pp. 281–297.

[2] T. Kanungo, D. M. Mount, N. S. Netanyahu, C. D. Piatko, R. Silverman, and A. Y. Wu, "An efficient k-means clustering algorithm: Analysis and implementation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 24, no. 7, pp. 881–892, Jul. 2002.

[3] J. B. Heaton, N. G. Polson, and J. H. Witte, "Deep learning in finance," *Annual Review of Financial Economics*, vol. 11, pp. 1–18, 2019.

[4] E. F. Fama, "Efficient capital markets: A review of theory and empirical work," *The Journal of Finance*, vol. 25, no. 2, pp. 383–417, 1970.

[5] J. J. Murphy, *Technical Analysis of the Financial Markets*. New York, NY, USA: New York Institute of Finance, 1999.

[6] F. Chollet, *Deep Learning with Python*. Shelter Island, NY, USA: Manning Publications, 2018.

[7] S. Raschka and V. Mirjalili, *Python Machine Learning*. Birmingham, U.K.: Packt Publishing, 2019.

[8] L. Kaufman and P. J. Rousseeuw, *Finding Groups in Data: An Introduction to Cluster Analysis*. Hoboken, NJ, USA: Wiley, 2005.

[9] D. Easley, M. López de Prado, and M. O'Hara, "The microstructure of the 'flash crash': Flow toxicity, liquidity crashes, and the probability of informed trading," *The Journal of Portfolio Management*, vol. 37, no. 2, pp. 118–128, 2011.

[10] Scikit-learn Developers, "KMeans," *Scikit-learn Documentation*. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html

[11] Scikit-learn Developers, "StandardScaler," *Scikit-learn Documentation*. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html

[12] S. Aghabozorgi, A. S. Shirkhorshidi, and T. Y. Wah, "Time-series clustering – a decade review," *Information Systems*, vol. 53, pp. 16–38, 2015.

[13] J. Patel, S. Shah, P. Thakkar, and K. Kotecha, "Predicting stock and stock price index movement using trend deterministic data preparation and machine learning techniques," *Expert Systems with Applications*, vol. 42, no. 1, pp. 259–268, 2015.

[14] S. Nanda, B. Mahanty, and M. K. Tiwari, "Clustering Indian stock market data for portfolio management," *Expert Systems with Applications*, vol. 37, no. 12, pp. 8793–8798, 2010.

[15] A. Tharwat, "Classification assessment methods," *Applied Computing and Informatics*, vol. 17, no. 1, pp. 168–192, 2020.

[16] Y. Jha, "Market Regime Discovery Code Repository," GitHub repository, 2026. [Online]. Available: https://github.com/Yashjha001/IIS_PROJECT